Shajith Sasikumar

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Education

Western University — London, ON
BESc, Electrical Engineering + Ivey HBA — *Expected Graduation: May 2028*

Work Experience

Software Developer Intern — Dec. 2025 – Feb. 2026
ThinkThamizh — *Toronto, ON*

- Shipped user-facing **React** features for the event signup flow, improving usability and accessibility and increasing completed signups by 10%+
- Engineered 10+ pages and interactive **JavaScript** components on the organization's **Shopify** site with custom **HTML** and **CSS**, shipped on a scheduled release cycle
- Built AI-assisted content automation workflows with **ChatGPT**, **Claude**, and **Cursor** to streamline digital media production, cutting time to publish by over 60%

Software Engineering Intern — Feb. 2026 – Apr. 2026
GreenVeil Packaging — *Ottawa, ON*

- Built a **Python** data pipeline that ingested pricing data from 5+ sources through **REST APIs**, cleaned and normalized thousands of records, and cut manual collection by over 80%
- Automated the ETL flow into multi-sheet **Excel** workbooks with monthly aggregations, standardizing multi-year pricing data for consistent reporting and trend analysis

Manufacturing Engineering Intern — Apr. 2026 – Aug. 2026
Visual Elements Manufacturing — *Woodbridge, ON*

- Built a **Python** CAD-to-CNC pipeline on the **SolidWorks API** that converted models into **woodWOP** programs and sent jobs to **HOMAG** CNCs over the network, improving workflow efficiency by 35%
- Interfaced with machine sensors and **PLC** controls and analyzed CNC status, run-time, and error data in **Excel VBA** dashboards, reducing production errors by 85%
- Modeled **SolidWorks** millwork fixtures for Lululemon, UGG, and Nordstrom and automated drawing exports and revision control with **Python** scripts, reviewing CNC output with engineers

Projects

Schematica | *React, TypeScript, Rust, WebAssembly, Python, PyTorch, FastAPI, PostgreSQL, Docker*

- Turned live circuit data into a navigable **React** and **TypeScript** interface with node voltages, branch currents, and waveform plots, from a photo of a hand-drawn schematic through **FastAPI**
- Trained a YOLO-based object detection model in **PyTorch** with an **OpenCV** wire-tracing and OCR pipeline to identify components, read their values, and automatically generate SPICE-style netlists
- Engineered a **Rust** circuit solver using Modified Nodal Analysis for DC and transient analysis, compiled to **WebAssembly** for real-time simulation, with LLM-generated step-by-step explanations

Marq Living | *Go, Python, TypeScript, Next.js, Supabase, TimescaleDB, MQTT, ESP32, C++*

- Built a full-stack building platform for a student residence with a **Next.js** and **Supabase** tenant app for live shuttle tracking, targeted announcements, front desk tickets, and amenity booking
- Engineered **Go** services that stream **ESP32** current-sensor data over **MQTT** into **TimescaleDB** with sub-second latency, plus a rules engine that turns bookings and occupancy into lighting and HVAC control
- Developed **Python** models on sensor time series for laundry state detection (95%+ accuracy) and 15-minute load forecasting (under 10% MAPE), plus EV scheduling that cut simulated peaks by 20%+

DevAssist | *Go, Python, TypeScript, FastAPI, React, PostgreSQL, Docker, Kafka, Redis*

- Built a **Go** indexer that clones repositories, chunks functions with tree-sitter, and stores embeddings in **PostgreSQL** (pgvector) for semantic search, reusing unchanged chunks on each reindex
- Engineered a **Go** sandbox that checks each patch in a locked-down **Docker** container (no network, non-root) with tests, security scans, and static analysis, and covered the **React** status view with tests

Skills

Languages: TypeScript, JavaScript, Python, Go, C, C++, Rust, Java, MATLAB, VBA
Frameworks & Tools: React, Next.js, FastAPI, Docker, PostgreSQL, Redis, Git, PyTorch
Hardware & Manufacturing: Sensors, ESP32, PLCs, CNC (HOMAG), woodWOP, SolidWorks, OnShape